PV101 Cohort 6 Day 3 - Instructor Packet

Wednesday 2026-07-08 | Rich teaching solo | Tim with trail crew | 6 students

Day 3 Snapshot

Item	Plan
Instructor	Rich leads the full classroom day.
Student count	6 students, run as 2 groups of 3 for lab.
Morning focus	Quiz and review for Solar Photovoltaics Basics by Shawn White, Chapters 3-4.
Afternoon focus	De-energized wire stripping and crimping lab.
Tone	Calm, slow, descriptive, first-time friendly.
Hard rule	No energized wiring practice. Practice wire and connectors only.

Learning Targets

- Explain the difference between sunlight as a resource and electricity as system behavior.
- Identify PV system parts from the Day 1 and Day 2 system path.
- Use power language correctly: volts, amps, watts, and energy over time.
- Prepare de-energized practice wire without damaging strands.
- Make a basic crimp connection and inspect it using a tug test and visual checklist.
- Work in a group of three using rotating roles: Tool Lead, Quality Inspector, Recorder.

Full Day Flow

Time	Block	Rich Notes
10:00-10:10	Arrival, reset, attendance	Rich sets tone: Tim is with trail crew, class runs normally.
10:10-10:30	Morning quiz	Chapters 3-4 reading check. Quiet individual work.
10:30-10:55	Quiz review	Students correct in a different color. Focus on ideas, not embarrassment.
10:55-11:20	Mini lesson: PV behavior	Sunlight, DC electricity, module behavior, shade, basic system flow.
11:20-11:30	Break	Reset tables for wiring tools.
11:30-12:00	Whiteboard: wire prep matters	Bad strip, good strip, connector choice, crimp quality, tug test.
12:00-12:30	Tool demo	Rich demonstrates de-energized wire stripping and crimping. Students observe first.

12:30-1:00	Lunch	Keep tools put away during lunch.
1:00-1:15	Lab setup	Two groups of three, roles assigned, safety check.
1:15-2:15	Lab: strip and crimp practice	Students complete practice pieces and inspection sheet.
2:15-2:35	Quality inspection	Groups swap samples and inspect for common errors.
2:35-2:50	Exit ticket	One skill they can do, one skill they need to keep practicing.
2:50-3:00	Cleanup	Tools counted, scraps sorted, benches cleared.

Rich Quick Start Script

Say this at the start:

> ***Today Tim is out with trail crew and I am running the room. We are going to start with a short reading quiz, review the big ideas, then spend the afternoon doing a practical lab: stripping wire and making clean crimp connections. The goal is not speed. The goal is clean work we can inspect.***

Morning Quiz Instructions

- Pass out quiz face-down.
- Give students 15-20 minutes.
- Allow book/notes only if that is your chosen class rule. Recommended: closed-book first, then correct with notes during review.
- Review question by question. Have students correct in a different color.
- Use missed questions as teaching feedback, not as a shame parade. Humanity tried that already; it was not impressive.

Mini Lesson: Chapters 3-4 Review Frame

Concept	Plain-language explanation	Board cue
Solar resource	Sunlight can be described as power right now or energy over time.	Irradiance = now. Insolation = over time.
PV cell/module/array	Cells combine into modules. Modules combine into arrays.	Cell -> Module -> Array
DC path	PV and batteries are DC side equipment.	PV -> Controller -> Battery -> DC Loads
Power	Watts describe work being done by electrical pressure and flow.	W = V x A
Verification	Labels help, but meters and inspection confirm.	Trust labels less than measured reality.

Afternoon Lab: Wire Stripping and Crimping

This lab is about muscle memory, visual inspection, and serviceable work. Everything is de-energized practice material.

Group	Roles	Rotation
Group A	Tool Lead, Quality Inspector, Recorder	Rotate after each completed connection.
Group B	Tool Lead, Quality Inspector, Recorder	Rotate after each completed connection.

Lab Safety Rules

- Use only wire and connectors Rich provides for practice.
- No students connect anything to batteries, solar panels, outlets, vehicles, or live equipment during this lab.
- Cut away from fingers and bodies.
- Eye protection recommended when clipping wire ends.
- All tools return to the tool count area before cleanup is complete.

Lab Demo Steps for Rich

1. Pick one piece of de-energized practice wire from the instructor-approved bin.
2. Identify the wire size if it is labeled. If not labeled, ask Rich before choosing the stripper slot.
3. Choose a connector that matches the wire size and lab objective.
4. Set the wire in the correct stripper hole. Close the tool gently, pull straight, and remove the insulation.
5. Inspect the exposed copper. Look for cut strands, crushed strands, too much exposed copper, or too little exposed copper.
6. Insert the wire fully into the connector barrel until it seats properly.
7. Place the connector in the correct crimper die and crimp once with firm, controlled pressure.
8. Perform a tug test by pulling the wire and connector in opposite directions with reasonable hand force.
9. Record whether the connection passed or failed and why.
10. Cut off failed practice connections and redo them as practice, not as punishment. The copper does not have feelings.

Quality Standard

Check	What passing looks like
Correct strip length	No bare copper sticking too far past connector; no insulation trapped under crimp.
No missing strands	Copper strands are present and not sliced off by the stripper.
Connector seated	Wire is inserted fully into the connector barrel.
Crimp location	Crimp is on the metal barrel, not on loose insulation.
Tug test	Connection does not pull apart under reasonable hand force.
Clean finish	No sharp copper hairs sticking out; connection is easy to inspect later.

Common Lab Corrections

Problem	What it means	Rich response
Too much copper exposed	Strip length is too long.	Show connector barrel length and redo one slowly.
Strands cut off	Wrong stripper hole or too much pressure.	Switch slot and demonstrate less force.
Wire pulls out	Bad crimp or wrong connector size.	Redo and compare before/after.
Insulation crimped instead of copper	Wire not inserted correctly.	Cut off and redo; inspect before crimp.
Student rushing	They are chasing speed instead of quality.	Make them say the inspection steps out loud.

Bridge to Day 4

- Clean connections are the beginning of reliable solar work.
- Tomorrow can move toward fuses, wire paths, loads, and RV system layout.
- The capstone depends on serviceable wiring: someone else should be able to inspect and understand the work later.